

RESEARCH ARTICLE

An assessment of Alberta's strategy for controlling mountain pine beetle outbreaks

Evan C. Johnson¹  | Mark A. Lewis^{2,3}

¹Mathematical and Statistical Sciences, University of Alberta, Edmonton, Alberta, Canada

²Department of Mathematics and Statistics, University of Victoria, Victoria, British Columbia, Canada

³Department of Biology, University of Victoria, Victoria, British Columbia, Canada

Correspondence

Evan C. Johnson

Email: ecjohns1@ualberta.ca

Funding information

Genome Alberta, Grant/Award Number: L20TF; Genome Canada, Grant/Award Number: 18202; Natural Sciences and Engineering Research Council of Canada, Grant/Award Number: RGPIN-2024-04792

Handling Editor: Bret Elderd

Abstract

1. Managing forest pests at landscape scales presents one of the greatest challenges in applied ecology. Since 2004, the Canadian province of Alberta spent more than 500 million dollars managing mountain pine beetle (MPB) populations—primarily by detecting and burning infested trees—yet the effectiveness of this intervention remains uncertain.
2. Using a statistical modelling framework coupled with long-term field data, we examined how direct control measures, severe winters and host-tree depletion shaped the trajectory of Alberta's MPB outbreak between 2009 and 2020.
3. Control efforts reduced cumulative tree mortality by 79% (95% predictive interval: 55%–89%), preventing approximately 1.8 (0.77–3.8) trees per hectare from being killed during the study period.
4. Cold winters had minimal direct impact on tree mortality, but worked synergistically with control efforts to collapse beetle populations around 2020. Host-tree depletion played a negligible role.
5. Each infestation that was removed (cost: 320 CAD) prevented the loss of approximately seven additional trees in the long term (95% predictive interval: 2.4–14), demonstrating potential cost-effectiveness. Model projections show high uncertainty in future outbreak severity, with potential tree mortality ranging from 0.37 to 8.6 trees per hectare over an 11-year period under a no-control scenario, and an alternative model suggesting even wider variation (~1–40 trees per hectare).
6. *Synthesis and applications.* Our findings tentatively support a 'wait it out' management strategy for forest pests with strong Allee effects, where small populations face experience negative growth. Moderate but sustained control efforts prevent explosive population growth until extreme winter conditions deliver a final blow, offering a cost-effective approach that enables complete outbreak suppression without indefinite intervention.

KEYWORDS

Allee effect, bioeconomics, Canada, direct control, integrated pest management, management program assessment, mountain pine beetle, spatiotemporal modelling

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Journal of Applied Ecology* published by John Wiley & Sons Ltd on behalf of British Ecological Society.

1 | INTRODUCTION

Forest insect pests represent a significant economic and ecological threat to global forest resources, with annual costs exceeding 70 billion USD (Bradshaw et al., 2016). The introduction of non-native insect species has been accelerated by globalization, while climate change is associated with increased pest outbreaks (Huang et al., 2011). Early management strategies focused on eradication through pesticide use, but this approach has largely been abandoned due to multiple concerns (Liebhold, 2012). These include the negative impacts of pesticides on human health and ecosystems, the potential for prolonging outbreaks by suppressing natural enemies and the tendency for locally eradicated populations to be quickly reseeded from surrounding areas (Pluess et al., 2012).

Forest pest management has shifted to a paradigm of integrated pest management, which emphasizes sustainable population suppression with monitoring, prevention and multiple control strategies. For example, the spread of the spongy moth has been slowed by 80% (Sharov & Liebhold, 1998) through a combination of pesticide treatments, semiochemical treatments to disrupt mating, surveillance based on a gridded network of pheromone traps, domestic quarantine regulations to stop the transport of moth eggs, and 'Don't move firewood' public awareness campaigns (Liebhold et al., 2021). In practice, current management approaches typically focus on slowing invasions and protecting high-value areas rather than suppressing regional outbreaks. The Canadian province of Alberta broke with this conservative approach when it responded to an invasion of the mountain pine beetle (MPB; *Dendroctonus ponderosae* Hopkins) by directly removing infested trees across more than 150,000 km² of pine forests.

Beginning around 2000, MPB populations grew across western North America as drought weakened tree defences and warmer winters reduced overwintering beetle mortality (Carroll, Régnière, et al., 2006). MPB expanded their geographic range into Alberta through multiple long-distance dispersal events from British Columbia (detailed history in Appendix S1; also see Brett et al., 2025). In response to this invasion, Alberta implemented three management strategies: Level 1 treatment (individual tree removal), Level 2 treatment (large-scale harvesting) and the healthy pine strategy (preemptive stand removal). This study focuses on Level 1 treatment because it was the most costly method in Alberta's portfolio, Level 2 treatment proved less effective, and the healthy pine strategy cannot be evaluated due to data limitations. Level 1 treatment involved helicopter surveys to locate *red-topped trees* (trees first infested the previous year), after which ground crews were deployed to search within 50-m radii for *green-attack trees*, that is newly infested trees seeded by nearby red-topped trees but not yet showing needle discoloration. Nearly all detected green-attack trees were removed using the 'cut-and-burn' method (Government of Alberta, 2016). Importantly, Alberta's monitoring data collection was designed from the outset to enable independent evaluation of control effectiveness, facilitating retrospective analyses of the type presented here.

The long-term effectiveness of pest control may depend critically on synergy between anthropogenic control and natural (environmental) control. MPB populations are particularly vulnerable to severe winter weather, with temperatures below -35°C causing significant mortality and near 100% mortality below -40°C (Régnière & Bentz, 2007; Wygant, 1940). Additionally, MPB populations collapse when densities fall below a critical threshold (i.e. a strong *Allee effect*), as small populations cannot overcome tree defences (Raffa & Berryman, 1983; Boone et al., 2011). Therefore, even moderately effective control efforts combined with periodic severe weather events could trigger population collapse by driving densities below the Allee threshold. This suggests a 'wait it out' strategy where short-term control efforts can achieve long-term suppression. This strategy is particularly relevant because real-world control faces practical constraints. In Alberta, approximately half of beetles establish themselves beyond the 50-m survey radii used by ground crews (Johnson, Brush, & Lewis, 2025), and resource constraints prevent crews from visiting all infested sites, resulting in the removal of only 50% of new infestations in managed areas (Carroll et al., 2017).

Alberta spent over 500 million Canadian dollars (CAD) on MPB control, yet the long-term effectiveness of these efforts is unclear. Although several studies have evaluated Alberta's strategy and contributed important findings (reviewed in Appendix S1), the quantitative predictions from these studies can be improved upon with respect to elements that ensure robustness: (1) transparent methods for model calibration, (2) in-depth analysis of factors affecting model uncertainty and (3) use of models of intermediate complexity that have adequate realism but avoid overcomplexity. Some models are too simple for reliable quantitative predictions, while others are overly complex with poorly justified parameters. Given the limitations of previous research, it is critical to undertake an independent assessment.

Here, we evaluate Alberta's MPB control program using an empirically calibrated statistical model coupled with counterfactual simulations. Our approach improves upon previous assessments through systematic uncertainty quantification and an intermediate model complexity that balances realism with parameter identifiability. We address four key management questions:

1. *What ended the outbreak?* We examine the relative importance of control efforts, cold winters and host depletion in terminating the MPB outbreak. While all three mechanisms have been proposed (Janice Cooke quoted in CBC News (2022)), their respective contributions remain unclear. Cold winters are known to cause population crashes (Safranyik & Linton, 1991), and control efforts have been successful at smaller spatial scales (Hopping & Mathers, 1945). Host depletion as an outbreak-ending mechanism is less likely given Alberta's low infestation densities, but still warrants investigation. We also explore potential synergistic effects between control efforts and cold events—specifically, whether control measures can 'buy time' until severe overwintering mortality occurs.

2. *How many trees did control efforts save?* We use simulations to compare control and no-control scenarios, assessing both the absolute number of trees saved and the percentage of total damage mitigated. These metrics provide measures of absolute and relative control effectiveness in the long term. Our analysis covers 2009–2020, a period bounded by two significant events: a major MPB immigration from British Columbia in 2009, which effectively reset Alberta's landscape of infestations, and the outbreak's approximate end in 2020.
3. *What range of control effectiveness should we expect in future outbreaks?* To address this question, we model potential future outbreaks by allowing winter temperatures and other environmental factors (represented as year effects) to vary from year-to-year. Given the large interannual fluctuations in MPB population densities, we anticipate a wide range of outcomes with respect to long-term control effectiveness. At one extreme, some future outbreaks might result in substantial tree mortality despite control efforts. On the other hand, moderate control measures could entirely suppress outbreaks.
4. *How does the cost-effectiveness of MPB control vary with control intensity?* Theory suggests that increasing control efforts could yield disproportionate benefits by pushing beetle populations below the Allee threshold, swiftly ending outbreaks and greatly reducing future expenses. We calculate the cost per hectare of level 1 treatment at different levels of control efficacy, defined as the proportion of total infestations removed within the managed areas. Additionally, we calculate the efficiency—measured here as trees saved per 100 CAD—under Alberta's actual control regime; this enables stakeholders to compare control costs against the economic and ecological benefits of retaining trees.

2 | METHODS

Using an empirically calibrated model of MPB dynamics, we simulate the Alberta outbreak under different scenarios, for example with and without level 1 treatment. While conceptually simple, this methodology has many moving parts, including multiple data sources, multiple model components, model justification and simulation procedures. Full details can be found in [Appendices S2](#) and [S3](#); here we offer a streamlined description.

2.1 | Overview and approach

Our analysis centers on counterfactual simulations where key factors are systematically removed or modified. We primarily manipulate three primary components—the intensity of Level 1 control, the occurrence of severe winters and the effects of host-tree depletion—to examine their individual and combined influences on salient features of Alberta's outbreak, specifically the population collapse around 2020 and the total number of trees killed over the 11-year period from 2010 to 2020. This approach enables causal inference

(conditional on the model, as is always the case). For example, if population densities never collapse when MPB control is turned off, we can conclude that control was necessary for ending the outbreak. If collapse only occurs when both control and severe winters are not removed, then these two factors are jointly necessary.

2.2 | Data sources

Infestation data comes from aerial *Heli-GPS surveys* and targeted ground surveys. Each September or October, helicopter crews identify clusters of *red-topped trees*—trees initially infested the previous year—and record the GPS coordinates for each location. Because MPB typically disperse over short distances (Safranyik et al., 1992), ground crews subsequently conduct searches within 50-m radii of the Heli-GPS coordinates to locate *green-attack trees*—newly infested trees that have not yet developed red needles. The vast majority of discovered green-attack trees are 'sanitized' by felling followed by either burning or chipping. These surveys produce high-quality data with minimal measurement errors: helicopter surveys achieve positional accuracy within ± 30 m (Government of Alberta, 2016) and accurate infestation counts within ± 10 trees for 92% of clusters (Nelson et al., 2006); and ground crews detect 98.5% of green-attack trees within surveyed areas.

Several additional data sources were incorporated. Pine volume estimates derive from Alberta's extended Vegetation Inventory, a standardized digital forest product based on aerial imagery, silvicultural records and field data collected by forestry companies. Photo interpreters delineate forested areas and estimate species composition, with conifer volume (units: m^3ha^{-1}) estimated using models incorporating stand height, age and species-specific site index equations. We derived total pine volume within each cell (units: m^3) by multiplying conifer volume by the proportion of conifers that are pines and the total forested hectares. Lastly, overwintering survival probabilities were predicted using the temperature data and the model of Régnière and Bentz (2007).

Data preparation involved converting point data to grid cells and filtering by location and time period. Point-based survey data and pine volume estimates from 2009 to 2020 were rasterized into 5×5 km cells. The total number of infested trees in each cell combines sanitized green-attack trees from the current year plus red-topped trees observed the following year. Analysis was restricted to cells where MPB infestations are ecologically feasible: those with pine volume exceeding $28,250 \text{ m}^3$ (only 2% of infestations in cells occur below this threshold) and pure lodgepole pine populations (excluding cells where jack pine ancestry exceeds 10% sensu Cullingham et al., 2012), as recent evidence suggests jack pine forests are less susceptible to MPB (Johnson, Musso, et al., 2025; Srivastava & Carroll, 2023). For model fitting, we further subsetted to the intersection of surveyed areas across years (Figure 1). Our simulations covered a broader spatial extent; however, when calculating summary statistics from simulated data, we subset to the smaller spatial extent to ensure fair comparisons with real data.

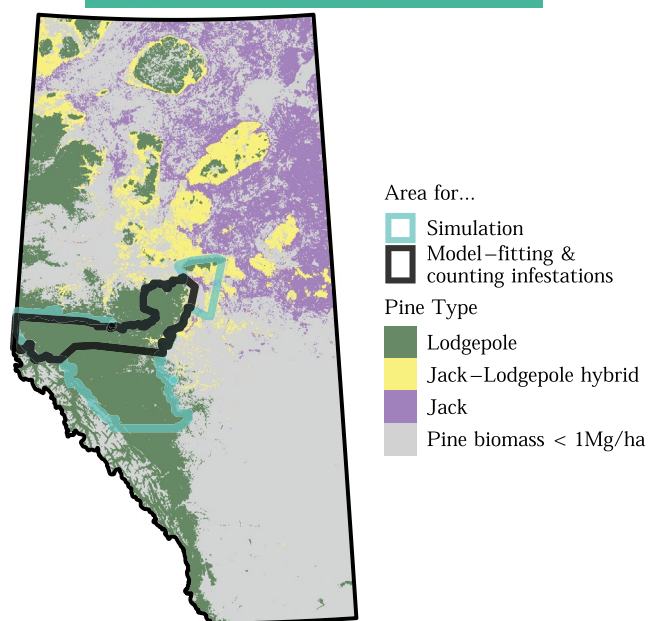


FIGURE 1 Areas in Alberta that are used for model fitting and simulations. The pine species data comes from Cullingham et al. (2012). Pine biomass data from Beaudoin et al. (2014) show the extent of pine across Alberta, since AVI-based pine volume estimates (used in the actual analysis) are not available for the entirety of the province.

2.3 | Model description

The model predicts beetle infestations via three life-history processes: overwintering survival based on weather data, dispersal following a fat-tailed Student's *t*-distribution, and local population dynamics captured by a zero-inflated negative binomial (ZINB) model. The ZINB structure reflects MPB biology: the zero-inflation captures both clustered dispersal and MPB's well-known *Allee effect* (Raffa & Berryman, 1983), where small populations fail to overcome tree defences, while the negative binomial component accounts for variable infestation counts due to clustered dispersal and other unmodeled factors (e.g., within-cell habitat heterogeneity). The model uses three predictors: beetle pressure (proportional to the expected number of attacking beetles after dispersal and overwintering mortality, calculated by multiplying the previous year's uncontrolled infestations with survival probabilities then convolving with a dispersal kernel), pine volume (as a measure of habitat suitability) and cumulative tree mortality (to test whether host-tree depletion via beetle attack influenced outbreak dynamics). The model also includes year effects to account for unexplained annual variation in beetle productivity and Gaussian processes to address residual spatial autocorrelation. Bayesian model fitting was implemented using *Stan* (Stan Development Team, 2020), with scalable Gaussian process algorithms adapted from Hoffmann and Onnela (2023).

The model uses a radially symmetric Student's *t*-distribution for dispersal, a fat-tailed distribution that allows for the occasional

long-distance dispersal. Specifically, 5% of beetles disperse further than 5 km, while the median dispersal distance is 50 m. The dispersal kernel parameters are fixed based on prior research (Johnson, Brush, & Lewis, 2025), but Appendix S7 shows that our conclusions are insensitive to different parameter values.

2.4 | Simulations

We conducted simulations across an extensive region in western Alberta, using observed 2009 infestations as initial conditions. Our approach centers on counterfactual scenarios that systematically manipulate key factors to isolate their individual and combined effects on outbreak dynamics. Counterfactual scenarios were constructed by manipulating model components: severe winters were removed by replacing spatiotemporally variable overwintering survival with a measure of central tendency (specifically the median of survival probability across 2009–2020 in all cells where at least one infestation occurred over the same period), host-tree depletion effects were removed by setting the cumulative killed trees predictor to zero, and control efforts were removed by setting control efficacy to zero.

Control efficacy was operationalized in simulations by allowing only a (1–control efficacy) fraction of infestations to contribute to the following year's beetle pressure calculations. For with-control simulations, the control efficacy parameter was set to 0.47 (the median efficacy observed from 2009 to 2020 in consistently surveyed areas), while no-control simulations used zero efficacy. Although real-world control efficacy varies with infestation density, proximity to untreated areas (Carroll et al., 2017) and management responses, we treat efficacy as a constant given our imperfect understanding of fine-scale MPB dynamics and the proprietary nature of Alberta's decision support system. Appendix S7 examines how results change under alternative efficacy values.

The simulations fall into two distinct categories: *actual outbreak* and *future outbreak* scenarios; this categorization is independent of other simulation variations (e.g., no-control vs. with-control). *Actual outbreak* scenarios use the specific environmental conditions observed from 2009 to 2019 (i.e. the exact sequence of winter temperatures and year effects) to understand what happened during Alberta's outbreak. *Future outbreak* scenarios use randomly generated environmental conditions—resampling winter temperature patterns from 2009 to 2019 and drawing year effects from their estimated distribution—to explore the range of outcomes possible under uncertain future conditions.

For each scenario, we conducted 1000 simulation runs, using a different posterior sample of model parameters for each simulation. This approach captures multiple sources of uncertainty: *parameter uncertainty* from the posterior distribution, *environmental stochasticity* from year effects and overwintering survival (but only in *future outbreak scenarios*) and *intrinsic stochasticity* arising from the Gaussian processes and negative binomial sampling.

2.5 | Management cost calculations

To find the cost of level 1 treatment, we use arithmetic, previously published figures and several assumptions. The cost of level 1 treatment, in units of CAD per tree, is calculated by dividing the total cost by the total number of controlled trees from 2004 to 2016, using figures from Hodge et al. (2017): 456,000,000 CAD/1,430,000 trees = 318.9 CAD per tree. To calculate the total cost at any hypothetical control efficacy level, we multiply the cumulative number of controlled trees from 2009 to 2019 by 318.9 CAD. We use the years 2009–2019 because control in these years influences infestation densities in the years which we aim to predict: 2010–2020. For better interpretability, we can calculate the 11-year cost per hectare by dividing the total cost by the 2,110,000 ha in our study area.

Several assumptions underpin these calculations. We exclude the cost of heli-GPS surveys, but this is unproblematic because these costs are negligible (0.15 CAD per hectare; Wulder et al., 2006) compared to ground survey and removal costs. A second unproblematic assumption is the linear relationship between cost and number of controlled trees, which is implied by the linear relationship between cost and ground survey area (Kunegel-Lion et al., 2019) and between ground survey area and number of controlled trees (Figure S4.1).

Two assumptions warrant more scrutiny. The first is the assumption of constant per-tree treatment costs, which only holds true near

the observed median control efficacy of 0.47—field crews are only sent to about 50% of red-topped tree sites, so more green-attack trees could be found simply by mobilizing more field crews. However, achieving higher control efficacy becomes increasingly expensive on a per-infestation basis. Once crews have addressed most red-topped tree sites, they must conduct wider concentric surveys at each site to attain higher efficacy. Expanding the survey radius from 50 to 300 m increases efficacy from 55% to 85%, but at 36 times the cost (Kunegel-Lion et al., 2019). The second questionable assumption is treating control efficacy as constant when it actually fluctuates. Given these limitations, our analysis serves best as a framework for order-of-magnitude cost estimates. Given the highly stochastic nature of MPB outbreaks, precise quantitative predictions are not a reasonable goal in any case.

3 | RESULTS

3.1 | What ended the outbreak?

A combination of control efforts and a sequence of severe winters ended the outbreak (Figure 2). When we simulated scenarios removing either control efforts or severe winters individually, the resulting infestation densities were approximately 10 times higher than

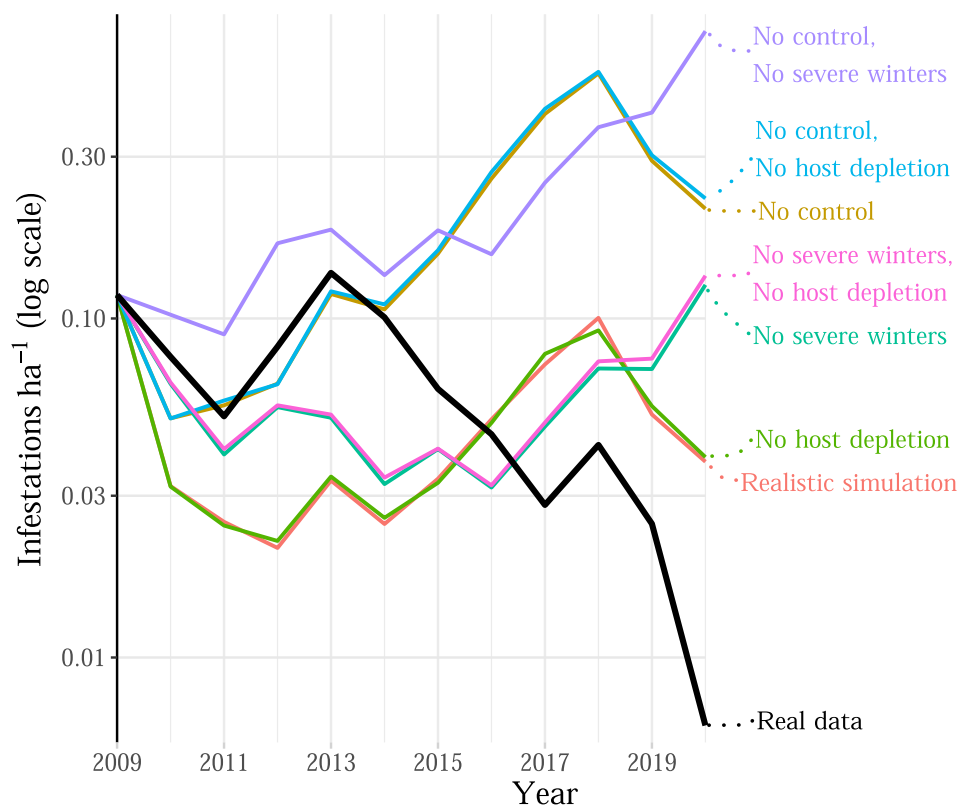


FIGURE 2 Time series of mountain pine beetle infestation densities, in both counterfactual simulation scenarios (coloured lines show the median value across simulations) and in the real data (black line). Simulations use the observed environmental parameters from 2009 to 2019, following the *actual outbreak* approach (see Section 2.4 for details). The y-axis values represent the spatial average of infestations ha^{-1} across the study area.

what was actually observed. Even more striking, simulations that removed both factors predicted infestation densities in 2020 that were approximately 100 times higher than observed levels. Host-tree depletion had no appreciable effect on outbreak trajectories, consistent with parameter estimates showing negligible effects of cumulative tree mortality compared to beetle pressure and pine volume (Appendix S5). The apparent discrepancy between the *Realistic simulation* line and the lower *Real data* line (Figure 2) has two explanations. First, simulations represent median values across multiple simulations while the real data show a single realization of a stochastic process. Second, the healthy pine strategy, which is not included in our model due to data limitations (Appendix S1), may have reduced actual infestation densities.

3.2 | What factors reduced the cumulative number of infestations?

Level 1 treatment is the primary factor limiting total infestations over the study period (Figure 3, Panel b). Although severe winters contributed to ending the outbreak, their impact on the cumulative number of infested trees was minimal, since most infestations

occurred between 2009 and 2018 before the onset of unusually cold winters.

3.3 | How many trees were saved by control efforts?

Over the 2009–2020 period, control efforts prevented the deaths of 1.8 trees per hectare, with a 95% posterior predictive interval (95% PI) of 0.77–3.8 trees per hectare (Table 1). This amounts to a 79% (95% PI: 55%–89%) reduction in tree deaths compared to a do-nothing scenario. Multiplied over the 2,110,000 total hectares of the consistently surveyed area (i.e. the black polygon in Figure 1), this translates to 3,798,000 trees saved over the 11-year period. Control effectiveness can also be expressed as the *control multiplier*, which shows that removing one tree through level 1 treatment prevents approximately 7 (95% PI: 2.4–14) infestations in the long run.

The *actual outbreak* simulations, which incorporate observed environmental conditions and level 1 treatment, produced a 95% predictive interval for cumulative infestations of 0.25–0.96 trees per hectare, with the observed value of 0.66 trees per hectare falling near the center of this range. This agreement between model

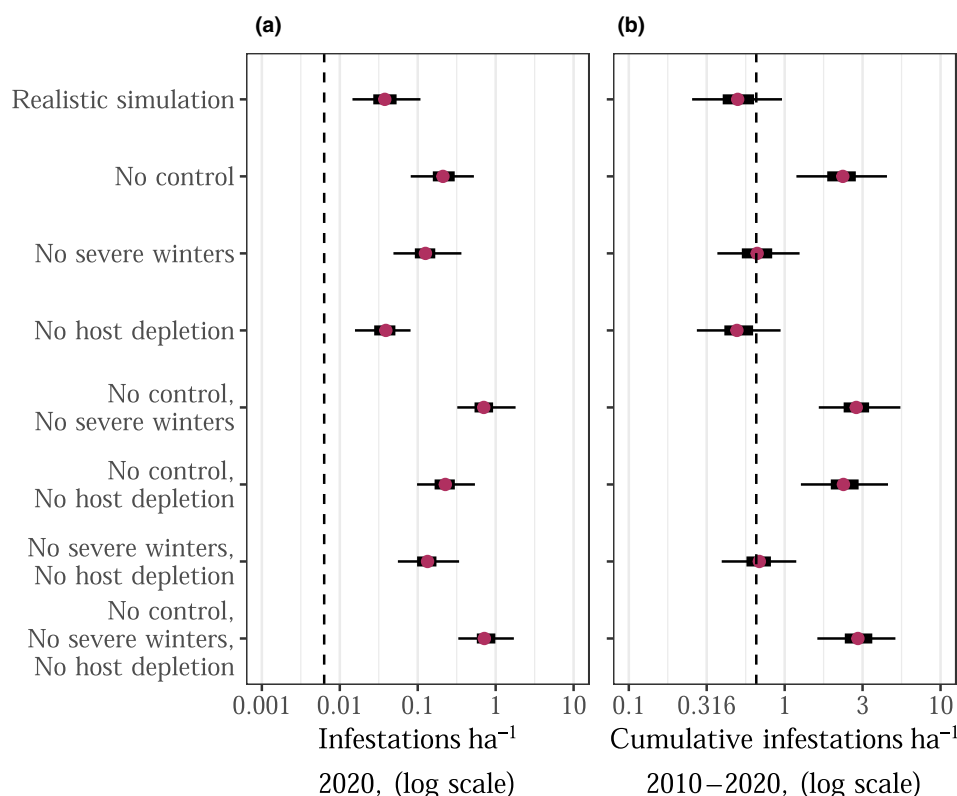


FIGURE 3 Outbreak quantities of interest (x-axis) across various simulation scenarios (y-axis). Panel (a): infestation densities at the end of the study period. Panel (b): cumulative infestation densities across 11 years. Points, horizontal thick lines and thin lines, respectively, show the mean, 50% interval and 95% interval of the predictive posterior distributions. All values are spatial averages across the study area. Dashed vertical lines show the actual value. Simulations use the observed environmental parameters from 2009 to 2019, following the *actual outbreak* approach (see Section 2.4 for details).

TABLE 1 How many trees did control efforts save over an 11-year period?

Simulation scenario	Trees killed ha ⁻¹ (no control)	Trees killed ha ⁻¹ (control)	No. trees ha ⁻¹ saved by control	% trees saved by control	Control multiplier
Actual outbreak, 2010–2020	2.4 (1.2–4.5)	0.5 (0.25–0.96)	1.8 (0.77–3.8)	79 (55–89)	6.9 (2.4–14)
Future outbreak, 11-year period	2.1 (0.37–8.6)	0.54 (0.11–2.7)	1.4 (0.23–6.6)	72 (50–88)	4.6 (1.5–13)

Note: No-control versus Control columns correspond to a control efficacy of 0 versus 0.47 control; units are trees per hectare, averaged across the study area. The format of table entries is Median (95% predictive intervals). The *actual outbreak* simulation scenario uses estimated values of overwintering survival and year effects. The *Future outbreak* scenario uses bootstrapped maps of overwintering survival and randomly generated year effects. The *control multiplier* is calculated as the number of trees saved by control from 2010 to 2020, divided by the number of controlled trees from 2009 to 2019.

predictions and actual outcomes is noteworthy, as it demonstrates the model's ability to accurately predict long-term system behaviour despite being parameterized using 1-year-ahead relationships. In simulations of the *future outbreak* scenario, the range of annual infestations (0.11–2.7 trees/ha/year) is consistent with aerial overview survey data from the western United States (0.13–2.8 trees/ha/year based on Meddens et al. (2012); MPB mortality in the western US averaged 0.13–2.8 trees/ha/year calculated as 404–5872 million total trees killed by all bark beetles during 1997–2010, $\times$ MPB percentages of 43%–63% $\div$ ~94 million hectares of western forest $\div$ 14 years). Infestation densities were an order of magnitude higher during the 2000s outbreak in British Columbia, but this can be attributed to the dense monoculture of mature lodgepole pine in central British Columbia, a feature not replicated in Alberta's sparser and more mixed forests. In British Columbia, aerial surveys identified large swaths with 15%–30% and >30% canopy damage (Westfall & Ebata, 2007), which translates to tens to hundreds of infested trees per hectare given lodgepole pine stand densities exceeding 1000 trees/ha (Meddens et al., 2011; Nigh et al., 2008).

3.4 | What range of treatment outcomes should be expected in future outbreaks?

In the future where environmental conditions are unknown, outbreak trajectories are highly variable (second row in Table 1). Under a no-control scenario, MPB may kill anywhere from 0.37 to 8.6 trees per hectare. Under the control scenario, MPB may kill anywhere from 0.11 to 2.7 trees per hectare. Importantly, the difference between the control and no-control scenarios under the observed environmental conditions (i.e. $2.4 - 0.50 \approx 1.9$) is smaller than the range of outcomes under a control scenario but unknown environmental conditions (i.e. $2.7 - 0.11 \approx 2.6$). Put another way, the effect of environmental stochasticity is comparable to the effect of control. This comparison suggests that future outbreaks could potentially result in significantly higher infestation densities even with control measures in place. However, such outcomes should not be interpreted as a failure of control methods. The relative effectiveness of control in the uncertain *future outbreak* scenario is similar to the relative effectiveness in the *actual outbreak* scenario (50%–88% and 55%–89%, respectively).

3.5 | How does MPB control cost vary with control intensity?

Figure 4 illustrates three key metrics of MPB control: cost (CAD ha⁻¹), effectiveness (trees saved ha⁻¹) and efficiency (saved trees per 100 CAD), across varying levels of control efficacy (proportion of all infestations controlled with level 1 treatment). For the sake of clarity, we explicitly distinguish between the meanings of similar-sounding 'E' words: efficacy measures immediate or controlled-setting results (operationalized here as the proportion of infested trees removed via level 1 treatment), effectiveness evaluates long-term impact, and efficiency is the quotient of outcomes to costs.

The relationship between control efficacy and cost is hump-shaped. Near the median observed value of 47% control efficacy (vertical line in Figure 4), costs range from 50 to 300 CAD per hectare, with this wide range reflecting uncertainty in future outbreak trajectories. More explosive outbreaks require treating more infested trees, increasing costs. The highest costs occur at intermediate control efficacy levels, where beetle populations continue growing despite control efforts. Both low and high control efficacies show lower costs, but for different reasons: low efficacy requires minimal intervention while high efficacy completely suppresses outbreaks, leading to fewer treatments in the long run.

Increasing control efficacy increases efficiency (measured as trees saved per 100 CAD), with higher efficacy levels yielding disproportionate benefits. This not only supports Carroll, Shore, et al. (2006) conclusion that early and intense control is most effective, but extends it by showing that such a strategy is also more efficient/cost-effective. The relationship between efficacy and efficiency is notably nonlinear, driven by the geometric growth of unchecked MPB populations. This is illustrated in Figure S4.2, where small decreases in control efficacy lead to exponential increases in infestation densities.

4 | DISCUSSION

Our analysis revealed several insights about the long-term effectiveness of Alberta's control strategy. Level 1 treatment reduced the total number of infestations by 79% (95% PI: 55%–89%) over an 11-year period and was the only factor that substantially reduced total damage. Neither cold

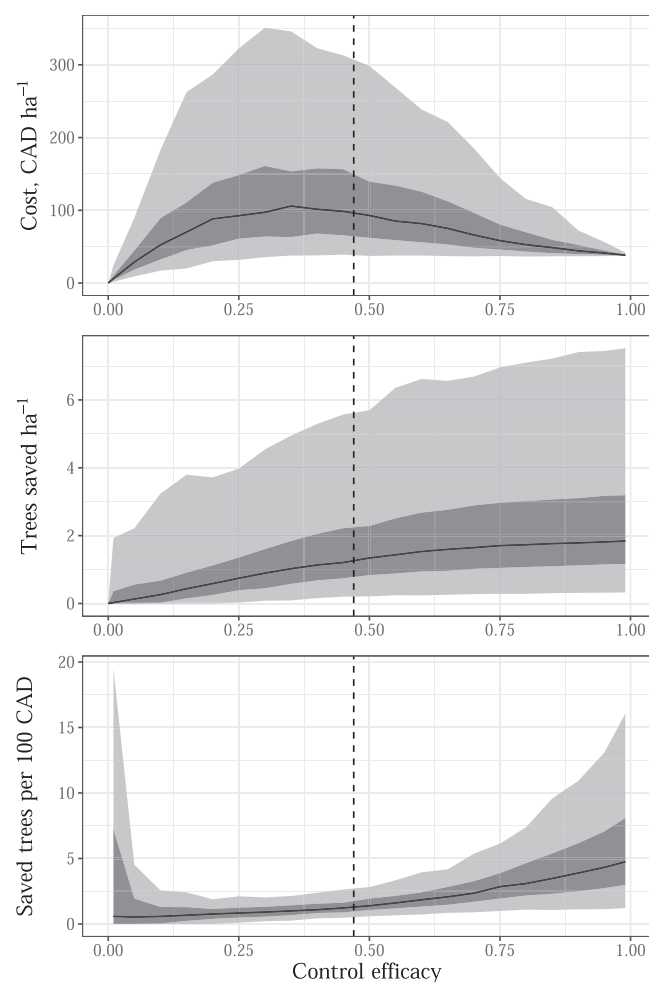


FIGURE 4 Cost, effectiveness (trees saved) and efficiency (trees saved per 100 CAD), as a function of control efficacy (proportion of infestations removed). Lines and grey ribbons show the median, 50% and 95% predictive intervals across *future outbreak* simulations. The vertical dashed line shows the actual control efficacy (median across 2009–2019). Y-axis values show the cumulative number of trees over an 11-year period (2009–2019 for controlled trees, 2010–2020 for saved trees), averaged over the study area. The actual cost was approximately 110 CAD per hectare for the 2009–2019 time period.

winters nor host-tree depletion (specifically from MPB, not logging) had a significant impact. However, cold winters and level 1 treatment were equally important in causing population collapse in the late 2010s. The trajectory of future outbreaks and control effectiveness is highly uncertain due to stochastic environmental conditions. This uncertainty also means that achieving the same control efficacy (approximately 50%) in future outbreaks could cost anywhere from 50 to 300 CAD per hectare. While higher control efficacy (the proportion of all infestations that are treated) would reduce costs by completely suppressing outbreaks, achieving high efficacy likely requires technological advances, in particular the remote sensing of early-infested green-attack trees.

Cold winters and Level 1 treatment acted synergistically in terminating the outbreak, lending some support to a ‘wait it out’

strategy, wherein moderate control effort is applied until extreme cold weather occurs. MPB population crashes due to extreme cold have been documented since the 1930s (Evenden & Gibson, 1940), but only two instances of collapse at a provincial/state scale have been recorded. The first was in British Columbia, where an emerging epidemic was stopped by a single winter in 1984–1985, with a cold snap in late October and minimum temperatures reaching -43°C (Safranyik & Linton, 1991). This paper establishes the Alberta collapse as a second example, with the notable distinction that the collapse occurred over a sequence of cold winters. Our analysis likely understates the impact of cold temperatures, since it only includes data through the winter of 2019 (beginning in December 2019), truncating the series of unusually cold winters that occurred from 2018 to 2023.

The trajectories of future MPB outbreaks are highly uncertain. This partially stems from fluctuating environmental factors (here overwintering mortality and year effects), with our main model showing that the cumulative killed trees may vary across an order of magnitude. Similarly, the absolute effectiveness of control—measured as the number of trees saved over an 11-year period—varies across an order of magnitude. Another source of uncertainty is model uncertainty, as demonstrated by an alternative model (Appendix S6) where uncontrolled outbreaks are eight times more destructive than in the main model. Further uncertainty stems from factors that are extremely difficult to model, including the initial conditions of future outbreaks, long-distance dispersal from British Columbia and the uncertain interaction between climate change and MPB dynamics (Bentz et al., 2022; Cooke & Carroll, 2017). It is also important to note that two modelling limitations have opposing biases on our estimates of control effectiveness: assuming constant control efficacy likely underestimates control benefits, while excluding massive immigration events likely overestimates them (Appendix S8). Readers should focus on orders of magnitude or relative metrics (i.e. proportion of trees saved) as these better reflect both modelling limitations and the inherent challenges of forecasting populations with highly stochastic dynamics.

Metrics of control effectiveness like ‘trees saved’ might seem short-sighted given Canada’s extensive, self-regenerating forests, but they serve as proxies for how control may mitigate the broader impacts of MPB outbreaks. MPB outbreaks cause economic harm (Corbett et al., 2016; Price et al., 2010) and have many ecological consequences, including shifts in forest structure from even-aged to uneven-aged stands, improved habitat for woodpeckers and saproxylic beetles and altered carbon dynamics (Dhar et al., 2016 and sources therein). Indeed, tree decomposition from the severe 2000s outbreak converted British Columbia forests from a carbon sink to a carbon source for at least 15 years (Kurz et al., 2008). Some ecological consequences may be viewed as beneficial, such as an increase in the diversity of understory plants and certain mammals following an outbreak (Pec et al., 2015; Saab et al., 2014). Finally, MPB outbreaks alter hydrological functioning in a variety of ways (Schnorbus, 2011). We explicitly do not make recommendations about whether control measures were worthwhile or how much should be invested in

future control efforts. Instead, our aim is to provide information to support public discourse and decision-making.

MPB is just one example of a bark beetle with landscape-scale impacts. The European spruce beetle affects outdoor recreation and soil erosion in northern Eurasia (Lecina-Diaz et al., 2024); and the red turpentine beetle impacts carbon sequestration and the timber industry in China (Sun et al., 2013). In an unusual but interesting case, the North American spruce beetle in Alaska's Kenai Peninsula actually increased property values by improving ocean views (Hansen & Naughton, 2013). These examples reflect a broader pattern: forest pest outbreaks have multifaceted but generally deleterious effects on economies and ecosystems, with these impacts increasing over time due to climate change (Bradshaw et al., 2016; Ji et al., 2011; Morris et al., 2018).

Notably, Alberta's MPB monitoring program was designed from the outset to enable independent evaluation of control effectiveness, facilitating the type of retrospective analysis presented here (Barry Cooke, *personal communication*). Our study builds on previous control assessments by evaluating multiple sources of uncertainty, including environmental stochasticity (i.e. climatic fluctuations), parameter uncertainty and model uncertainty. Additionally, our model is simpler than previous models in order to balance realism with additional goals: avoiding overfitting, avoiding manual parameter tuning and justifying model structure (Appendix S3) with the hope of reducing model bias. Despite substantially different modelling approaches, our results align with those of Carroll et al. (2017): we find that control measures save 55%–89% of trees, comparable to their finding of approximately 66% saved pine volume. This agreement can be verified by examining figure 9 in Carroll et al. (2017), specifically comparing the StS (for 'slow the spread') and StS ends 2020 scenarios from the 2020 to 2030 time period.

What comes next for MPB in Alberta? Since MPB affected only a small fraction of lodgepole pine, outbreaks could resurge within decades, though this depends on the yet-to-be-assessed effectiveness of Alberta's healthy pine strategy. Unlike British Columbia's widespread, episodic outbreaks that resulted from fire suppression and extensive lodgepole pine monocultures, Alberta's mixed forests may experience smaller, asynchronous outbreaks like those observed in the western United States from 1950 to 2000 (Axelson et al., 2018). Given that climate change and warming temperatures are expected to increase outbreak frequency (Brush & Lewis, 2023), our *future outbreak* simulations likely provide conservative estimates of infestation numbers. (Brush & Lewis, 2023), so our simulations of *future outbreaks* may be conservative in the number of infestations caused. Low-elevation MPB populations might suffer from accelerated development that disrupts the timing of their life cycle (i.e. fractional voltinism; Bentz et al., 2022). The beetle's eastward expansion through Alberta critically depends on its adaptation to jack pine, a novel host tree. Experimental evidence suggests jack pine is suitable for MPB development (e.g., Cerezke, 1995), but MPB may struggle to detect and coordinate mass attacks due to the jack pine's smaller size and distinct chemical profile (Johnson, Musso, et al., 2025).

Additionally, competition from wood-boring beetles in jack pine forests (Pokorný, 2021) may prevent MPB from establishing endemic populations in eastern Alberta.

Our assessment demonstrates important theoretical concepts in pest management. The dearth of MPB outbreaks since the 2020 collapse suggests that Allee effects can indeed be exploited for management purposes. While previous studies have stressed the theoretical importance of Allee effects in pest management (Tobin et al., 2011), demonstrated examples are rare (but see Liebhold & Tobin, 2010). Climate variability (here, severe weather) worked synergistically with control efforts to achieve outbreak suppression, adding important nuance to the baseline expectation that climate change will exacerbate pest problems. Integrated pest management emphasizes multiple control strategies, but our analysis was restricted to level 1 treatment. Future research should evaluate Alberta's healthy pine strategy (i.e. preventative clear-cutting of susceptible forests).

In Alberta, consistent control efforts reduced total pine mortality by roughly three-quarters over 11 years, demonstrating humans' capacity to meaningfully influence forest pest dynamics. Achieving rapid suppression would require both technological advances—the remote sensing of early-infested green-attack trees—and substantial investments, which may become increasingly difficult to secure amid the rise of anti-science political movements in the United States and Canada. The cost-effectiveness of MPB control (trees saved per dollar) increases as more money is spent, but there are no guarantees. Even with simplifying assumptions (e.g., no dispersal from British Columbia, no climate change effects), the trajectory of the next outbreak is highly uncertain, as is the absolute effectiveness of control. Our projections show future damage ranging from 0.11 to 2.7 trees/ha—a span of outcomes larger than the estimated effect of control in the most recent outbreak (1.8 trees/ha saved). These findings demonstrate that human intervention in complex ecological systems holds both great potential and great uncertainty.

AUTHOR CONTRIBUTIONS

Evan C. Johnson conceived of the study, performed the analysis and wrote the first draft; Evan C. Johnson and Mark A. Lewis contributed critically to the drafts and gave final approval for publication.

ACKNOWLEDGEMENTS

The authors would like to thank Allan Carroll, Barry Cooke, Kathy Bleiker, Antonia Musso, Micah Brush, Kévan Rastello and Xiaoqi Xie for helpful discussions. We also thank two reviewers for their constructive comments and suggestions. Funding for this research has been provided through grants to the TRIA-FoR Project to MAL from Genome Canada (Project No. 18202) and the Government of Alberta through Genome Alberta (Grant No. L20TF), with contributions from the University of Alberta. MAL is grateful for the Betty and Gilbert Kennedy Chair in Mathematical Biology and for funding from the NSERC Discovery Grant program (RGPIN-2024-04792). ECJ thanks Logan Brissette for helping to design and create the graphical abstract.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Code and data are available on Figshare, <https://doi.org/10.6084/m9.figshare.30073387> (Johnson, 2025).

ORCID

Evan C. Johnson  <https://orcid.org/0000-0001-6352-2584>

REFERENCES

- Axelsson, J. N., Hawkes, B. C., van Akker, L., & Alfaro, R. I. (2018). Stand dynamics and the mountain pine beetle—30 years of forest change in Waterton Lakes National Park, Alberta, Canada. *Canadian Journal of Forest Research*, 48(10), 1159–1170.
- Beaudoin, A., Bernier, P., Guindon, L., Villemaire, P., Guo, X., Stinson, G., Bergeron, T., Magnussen, S., & Hall, R. (2014). Mapping attributes of Canada's forests at moderate resolution through k NN and MODIS imagery. *Canadian Journal of Forest Research*, 44(5), 521–532.
- Bentz, B. J., Hansen, E. M., Davenport, M., & Soderberg, D. (2022). Complexities in predicting mountain pine beetle and spruce beetle response to climate change. In *Bark beetle management, ecology, and climate change* (pp. 31–54). Elsevier.
- Boone, C. K., Aukema, B. H., Bohlmann, J., Carroll, A. L., & Raffa, K. F. (2011). Efficacy of tree defense physiology varies with bark beetle population density: A basis for positive feedback in eruptive species. *Canadian Journal of Forest Research*, 41(6), 1174–1188.
- Bradshaw, C. J., Leroy, B., Bellard, C., Roiz, D., Albert, C., Fournier, A., Barbet-Massin, M., Salles, J.-M., Simard, F., & Courchamp, F. (2016). Massive yet grossly underestimated global costs of invasive insects. *Nature Communications*, 7(1), 12986.
- Brett, R. D., Volney, J. A., Cerezke, H. F., Cooke, B. J., & Weber, J. D. (2025). *A history of mountain pine beetle outbreaks in Alberta and Saskatchewan, 1940–2007*. Information Report GLC-X-39, Natural Resources Canada, Canadian Forest Service, Great Lakes Forestry Centre.
- Brush, M., & Lewis, M. A. (2023). Coupling mountain pine beetle and forest population dynamics predicts transient outbreaks that are likely to increase in number with climate change. *Bulletin of Mathematical Biology*, 85(11), 108.
- Carroll, A., Régnière, J., Logan, J., Taylor, S., Bentz, B., & Powell, J. (2006). *Impacts of climate change on range expansion by the mountain pine beetle*. Technical report, Pacific Forestry Centre Canada.
- Carroll, A., Seely, B., Welham, C., & Nelson, H. (2017). *Assessing the effectiveness of Alberta's forest management program against the mountain pine beetle: Final report for fRI research project 246.18 parts 1 and 2*. Technical report, fRI Research.
- Carroll, A. L., Shore, T. L., & Safranyik, L. (2006). Direct control: Theory and practice. In L. Safranyik & B. Wilson (Eds.), *Detection, mapping, and monitoring of the mountain pine beetle* (pp. 155–172). Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre.
- CBC News. (2022). *Mountain pine beetle populations down by 94 per cent in Alberta since 2019: Province*.
- Cerezke, H. (1995). Egg gallery, brood production, and adult characteristics of mountain pine beetle, *Dendroctonus ponderosae* Hopkins (Coleoptera: Scolytidae), in three pine hosts. *The Canadian Entomologist*, 127(6), 955–965.
- Cooke, B. J., & Carroll, A. L. (2017). Predicting the risk of mountain pine beetle spread to eastern pine forests: Considering uncertainty in uncertain times. *Forest Ecology and Management*, 396, 11–25.
- Corbett, L. J., Withey, P., Lantz, V., & Ochudho, T. (2016). The economic impact of the mountain pine beetle infestation in British Columbia: Provincial estimates from a CGE analysis. *Forestry: An International Journal of Forest Research*, 89(1), 100–105.
- Cullingham, C. I., James, P. M., Cooke, J. E., & Coltman, D. W. (2012). Characterizing the physical and genetic structure of the lodgepole pine × jack pine hybrid zone: Mosaic structure and differential introgression. *Evolutionary Applications*, 5(8), 879–891.
- Dhar, A., Parrott, L., & Hawkins, C. D. (2016). Aftermath of mountain pine beetle outbreak in British Columbia: Stand dynamics, management response and ecosystem resilience. *Forests*, 7(8), 171.
- Evenden, J. C., & Gibson, A. L. (1940). A destructive infestation in lodgepole pine stands by the mountain pine beetle. *Journal of Forestry*, 38(3), 271–275.
- Government of Alberta. (2016). *Mountain pine beetle detection and management in Alberta*. Technical report, Government of Alberta, Agriculture and Forestry.
- Hansen, W. D., & Naughton, H. T. (2013). The effects of a spruce bark beetle outbreak and wildfires on property values in the wildland–urban interface of south-central Alaska, USA. *Ecological Economics*, 96, 141–154.
- Hodge, J., Cooke, B., & McIntosh, R. (2017). *A strategic approach to slow the spread of mountain pine beetle across Canada*. Technical report, Forest Pest Working Group of the Canadian Council of Forest Ministers.
- Hoffmann, T., & Onnela, J.-P. (2023). Scalable Gaussian process inference with Stan. *arXiv preprint*, arXiv:2301.08836.
- Hopping, G. R., & Mathers, W. (1945). Observations on outbreaks and control of the mountain pine beetle in the lodge-pole pine stands of Western Canada. *The Forestry Chronicle*, 21(2), 98–108.
- Huang, D., Haack, R. A., & Zhang, R. (2011). Does global warming increase establishment rates of invasive alien species? A centennial time series analysis. *PLoS One*, 6(9), e24733.
- Ji, L., Wang, Z., Wang, X., & An, L. (2011). Forest insect pest management and forest management in China: An overview. *Environmental Management*, 48(6), 1107–1121.
- Johnson, E. C. (2025). Data and code: “An assessment of Alberta's strategy for controlling mountain pine beetle outbreaks”. *figshare*. <https://doi.org/10.6084/m9.figshare.30073387>
- Johnson, E. C., Brush, M., & Lewis, M. A. (2025). Modeling stratified dispersal in forest pests: A case study of the mountain pine beetle in Alberta. *Ecology*, in press.
- Johnson, E. C., Musso, A., Cullingham, C., & Lewis, M. A. (2025). Biological barriers to forest pest invasions: A novel host tree slows mountain pine beetle range expansion. *Ecology and Evolution*, 15(10), e72296.
- Kunegel-Lion, M., McIntosh, R. L., & Lewis, M. A. (2019). Management assessment of mountain pine beetle infestation in Cypress Hills, SK. *Canadian Journal of Forest Research*, 49(2), 154–163.
- Kurz, W. A., Dymond, C., Stinson, G., Rampley, G., Neilson, E., Carroll, A., Ebata, T., & Safranyik, L. (2008). Mountain pine beetle and forest carbon feedback to climate change. *Nature*, 452(7190), 987–990.
- Lecina-Diaz, J., Senf, C., Grünig, M., & Seidl, R. (2024). Ecosystem services at risk from disturbance in Europe's forests. *Global Change Biology*, 30(3), e17242.
- Liebholt, A. M. (2012). Forest pest management in a changing world. *International Journal of Pest Management*, 58(3), 289–295.
- Liebholt, A. M., Leonard, D., Marra, J. L., & Pfister, S. E. (2021). Area-wide management of invading gypsy moth (*Lymantria dispar*) populations in the USA. In J. Hendrichs, R. Pereira, & M. J. B. Vervysen (Eds.), *Area-wide integrated pest management* (pp. 551–560). CRC Press.
- Liebholt, A. M., & Tobin, P. C. (2010). Exploiting the Achilles heels of pest invasions: Allee effects, stratified dispersal and management of forest insect establishment and spread. *New Zealand Journal of Forest Science*, 40, S25–S33.
- Meddens, A. J., Hicke, J. A., & Ferguson, C. A. (2012). Spatiotemporal patterns of observed bark beetle-caused tree mortality in British Columbia and the western United States. *Ecological Applications*, 22(7), 1876–1891.

- Meddens, A. J., Hicke, J. A., & Vierling, L. A. (2011). Evaluating the potential of multispectral imagery to map multiple stages of tree mortality. *Remote Sensing of Environment*, 115(7), 1632–1642.
- Morris, J. L., Cottrell, S., Fettig, C. J., DeRose, R. J., Mattor, K. M., Carter, V. A., Clear, J., Clement, J., Hansen, W. D., Hicke, J. A., Higuera, P. E., Seddon, A. W. R., Seppä, H., Sherriff, R. L., Stednick, J. D., & Seybold, S. J. (2018). Bark beetles as agents of change in social–ecological systems. *Frontiers in Ecology and the Environment*, 16(5), S34–S43.
- Nelson, T., Boots, B., & Wulder, M. A. (2006). Large-area mountain pine beetle infestations: Spatial data representation and accuracy. *The Forestry Chronicle*, 82(2), 243–252.
- Nigh, G. D., Antos, J. A., & Parish, R. (2008). Density and distribution of advance regeneration in mountain pine beetle killed lodgepole pine stands of the Montane Spruce Zone of southern British Columbia. *Canadian Journal of Forest Research*, 38(11), 2826–2836.
- Pec, G. J., Karst, J., Sywenky, A. N., Cigan, P. W., Erbilgin, N., Simard, S. W., & Cahill, J. F., Jr. (2015). Rapid increases in forest understory diversity and productivity following a mountain pine beetle (*Dendroctonus ponderosae*) outbreak in pine forests. *PLoS One*, 10(4), e0124691.
- Pluess, T., Cannon, R., Jarošík, V., Pergl, J., Pyšek, P., & Bacher, S. (2012). When are eradication campaigns successful? A test of common assumptions. *Biological Invasions*, 14(7), 1365–1378.
- Pokorny, S. W. (2021). *Novel biological interactions influence the persistence potential of invasive mountain pine beetle (Dendroctonus ponderosae) in the Canadian Boreal Forest* [PhD thesis]. University of British Columbia.
- Price, J. I., McCollum, D. W., & Berrens, R. P. (2010). Insect infestation and residential property values: A hedonic analysis of the mountain pine beetle epidemic. *Forest Policy and Economics*, 12(6), 415–422.
- Raffa, K., & Berryman, A. (1983). The role of host plant resistance in the colonization behavior and ecology of bark beetles (Coleoptera: Scolytidae). *Ecological Monographs*, 53(1), 27–49.
- Régnière, J., & Bentz, B. (2007). Modeling cold tolerance in the mountain pine beetle, *Dendroctonus ponderosae*. *Journal of Insect Physiology*, 53(6), 559–572.
- Saab, V. A., Latif, Q. S., Rowland, M. M., Johnson, T. N., Chalfoun, A. D., Buskirk, S. W., Heyward, J. E., & Dresser, M. A. (2014). Ecological consequences of mountain pine beetle outbreaks for wildlife in western North American forests. *Forest Science*, 60(3), 539–559.
- Safranyik, L., & Linton, D. (1991). Unseasonably low fall and winter temperatures affecting mountain pine beetle and pine engraver beetle populations and damage in the British Columbia Chilcotin Region. *Journal of the Entomological Society of British Columbia*, 88, 17–21.
- Safranyik, L., Linton, D., Silversides, R., & McMullen, L. (1992). Dispersal of released mountain pine beetles under the canopy of a mature lodgepole pine stand. *Journal of Applied Entomology*, 113(1–5), 441–450.
- Schnorbus, M. (2011). *A synthesis of the hydrological consequences of large-scale mountain pine beetle disturbance*. Technical report, Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre.
- Sharov, A. A., & Liebhold, A. M. (1998). Model of slowing the spread of gypsy moth (Lepidoptera: Lymantriidae) with a barrier zone. *Ecological Applications*, 8(4), 1170–1179.
- Srivastava, V., & Carroll, A. L. (2023). Dynamic distribution modelling using a native invasive species, the mountain pine beetle. *Ecological Modelling*, 482, 110409.
- Stan Development Team. (2020). *RStan: the R interface to Stan*. R package version 2.19.3.
- Sun, J., Lu, M., Gillette, N. E., & Wingfield, M. J. (2013). Red turpentine beetle: Innocuous native becomes invasive tree killer in China. *Annual Review of Entomology*, 58, 293–311.
- Tobin, P. C., Berec, L., & Liebhold, A. M. (2011). Exploiting Allee effects for managing biological invasions. *Ecology Letters*, 14(6), 615–624.
- Westfall, J., & Ebata, T. (2007). *2007 summary of forest health conditions in British Columbia*. Technical report, British Columbia Ministry of Forests and Range.
- Wulder, M. A., White, J., Bentz, B., & Ebata, T. (2006). Augmenting the existing survey hierarchy for mountain pine beetle red-attack damage with satellite remotely sensed data. *Forestry Chronicle*, 82(2), 187–202.
- Wygant, N. D. (1940). *Effects of low temperatures on the Black Hills beetle (Dendroctonus ponderosae Hopk.)* [PhD thesis], New York State College of Forestry.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. The history of MPB in Alberta.

Appendix S2. Supplementary methods.

Appendix S3. Model justification.

Appendix S4. Additional figures and tables.

Appendix S5. Supplementary results.

Appendix S6. Alternative models.

Appendix S7. Sensitivity analysis.

Appendix S8. Limitations.

How to cite this article: Johnson, E. C., & Lewis, M. A. (2025).

An assessment of Alberta's strategy for controlling mountain pine beetle outbreaks. *Journal of Applied Ecology*, 62, 3005–3015. <https://doi.org/10.1111/1365-2664.70178>